

Problem Chosen

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Summary Sheet

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Summary

Keywords:

Contents

1	Introduction	4
1.1	Problem Background	4
1.2	Restatement of the Problem	4
2		4
2.1		4
2.2	Notations	4
3	Estimating Latent Fan Votes: A Bayesian Inference Approach	4
3.1	Model Formulation and Notation	5
3.2	The Dirichlet Prior	5
3.3	Likelihood and Voting Constraints	5
3.3.1	Percentage-Based System (Seasons 3–27)	5
3.3.2	Rank-Based System (Seasons 1–2, 28–34)	5
3.4	Posterior Inference via MCMC	5
3.5	Measures of Consistency and Certainty	6
3.6		6
3.6.1		6
3.7	Differential Equation Model Based on System Dynamics	6
3.8		6
3.8.1	Shannon Diversity Index	6
4	Advanced Mathematical Framework for Factor Analysis	6
4.1	Phase I: Hierarchical Variance Decomposition (LMM)	6
4.1.1	Stochastic Formulation	7
4.1.2	Estimation via Restricted Maximum Likelihood (REML)	7
4.2	Phase II: Non-linear Perception and Game-Theoretic Attribution (SHAP)	7
4.2.1	Axiomatic Feature Attribution	7
4.2.2	Divergence Analysis: The Perception Gap	8
4.3	Phase III: Causal Path Integration (SEM)	8
4.3.1	Structural and Measurement Equations	8
4.4	The Unified Latent Achievement Model (ULAM)	8
5	Empirical Analysis and Results	9
5.1	The "Pro-Dancer Effect": Technical Coach or Fan Magnet?	9
5.2	The Perception Gap: Judges vs. Fans	9
5.3	Case Study Validation: The ULAM Score in Action	10
6	Conclusion on Factor Impact	11
7		11
7.1	Model Extension	11
7.2		11
7.2.1		11
7.2.2		11
7.2.3		11
7.3		12

8 Strengths and Weaknesses	12
8.1 Strengths	12
8.2 Weaknesses	12
References	12
Letter	13
Report on Use of AI	14

1 Introduction

1.1 Problem Background

1.2 Restatement of the Problem

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2

2.1

1. 11
2. 22
3. 33

2.2 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
E	entropy

*There are some variables that are not listed here and will be discussed in detail in each section.

3 Estimating Latent Fan Votes: A Bayesian Inference Approach

The primary challenge in analyzing "Dancing with the Stars" (DWTS) lies in the fact that fan votes are latent variables, never publicly disclosed by the producers. However, the weekly elimination results and judges' scores provide a set of observable constraints. We propose a Bayesian framework to reverse-engineer the distribution of fan vote proportions.

3.1 Model Formulation and Notation

Let k_t be the number of contestants in week t . We define the following variables:

- $\mathbf{S}_t = \{s_{1,t}, s_{2,t}, \dots, s_{k_t,t}\}$: The set of normalized judges' scores for each contestant.
- $\boldsymbol{\theta}_t = \{\theta_{1,t}, \theta_{2,t}, \dots, \theta_{k_t,t}\}$: The latent proportion of fan votes, where $\sum_{i=1}^{k_t} \theta_{i,t} = 1$ and $\theta_{i,t} \geq 0$.
- y_t : The observed elimination result (the index of the contestant who was eliminated).

3.2 The Dirichlet Prior

The fan vote proportions reside on a $(k_t - 1)$ -dimensional simplex. The Dirichlet distribution is the natural conjugate prior for such compositional data. We assume:

$$\boldsymbol{\theta}_t \sim \text{Dirichlet}(\alpha_{1,t}, \alpha_{2,t}, \dots, \alpha_{k_t,t}) \quad (1)$$

where $\alpha_{i,t}$ represents the "prior popularity" of contestant i . In the absence of specific demographic data, we initially set $\alpha_{i,t} = \alpha_0 = 1$ (a flat prior). For iterative estimation across weeks, we update $\alpha_{i,t}$ based on the posterior mean of week $t - 1$, adjusted for recent performance.

3.3 Likelihood and Voting Constraints

The likelihood function $L(y_t | \boldsymbol{\theta}_t, \mathbf{S}_t)$ is defined by the specific voting rules of the season. Unlike traditional statistical models, the "observation" here is a ranking outcome, which acts as a hard constraint on the parameter space.

3.3.1 Percentage-Based System (Seasons 3–27)

In this system, the total score $V_{i,t}$ is the sum of percentages. Let $P_{i,t}^J = s_{i,t} / \sum s_{j,t}$ be the judges' score percentage. The elimination condition for contestant j is:

$$L(y_t = j | \boldsymbol{\theta}_t, \mathbf{S}_t) = \begin{cases} 1, & \text{if } P_{j,t}^J + \theta_{j,t} < P_{i,t}^J + \theta_{i,t} \quad \forall i \neq j \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

3.3.2 Rank-Based System (Seasons 1–2, 28–34)

In the rank-based system, let $R_{i,t}^J$ be the rank of contestant i based on judges' scores and $R_{i,t}^F$ be the rank based on fan votes $\theta_{i,t}$. The contestant with the highest sum of ranks (lowest performance) is eliminated:

$$L(y_t = j | \boldsymbol{\theta}_t, \mathbf{S}_t) = \begin{cases} 1, & \text{if } R_{j,t}^J + R_{j,t}^F > R_{i,t}^J + R_{i,t}^F \quad \forall i \neq j \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

3.4 Posterior Inference via MCMC

The posterior distribution $P(\boldsymbol{\theta}_t | y_t, \mathbf{S}_t) \propto L(y_t | \boldsymbol{\theta}_t, \mathbf{S}_t) \cdot \pi(\boldsymbol{\theta}_t)$ is analytically intractable due to the discontinuous nature of the likelihood. We employ Markov Chain Monte

Carlo (MCMC) sampling, specifically the Metropolis-Hastings algorithm, to sample from the constrained simplex space.

By generating $N = 10,000$ samples for each week, we estimate the mean fan vote $\hat{\theta}_{i,t} = E[\theta_{i,t}|\text{data}]$ and the associated uncertainty.

3.5 Measures of Consistency and Certainty

To evaluate the model, we define two metrics:

1. **Consistency Index (C):** The proportion of weeks where the maximum a posteriori (MAP) estimate of θ_t correctly identifies the eliminated contestant y_t .
2. **Certainty Score (σ):** Defined as the inverse of the Shannon Entropy of the posterior distribution:

$$\text{Certainty} = \left(- \sum_{i=1}^{k_t} \hat{\theta}_{i,t} \ln \hat{\theta}_{i,t} \right)^{-1} \quad (4)$$

High certainty indicates that the judges' scores were so disparate that only a very specific range of fan votes could have caused the observed outcome.

3.6

3.6.1

3.7 Differential Equation Model Based on System Dynamics

The overall model can be expressed as:

3.8

3.8.1 Shannon Diversity Index

4 Integrated Mathematical Framework and Empirical Analysis

4.1 Advanced Mathematical Framework for Factor Analysis

The evaluation of success in *Dancing with the Stars* (DWTS) is characterized by a hierarchical structure where celebrity potential is moderated by professional expertise. We derive an integrated model through three stages of mathematical refinement to quantify how celebrity traits and professional partners impact judges' scores and fan votes differently.

4.1.1 Phase I: Hierarchical Variance Decomposition (LMM)

To isolate the "Pro-Dancer Effect," we acknowledge that contestants are nested within professional partners. A standard regression ignores this clustering, leading to underestimated standard errors. We utilize a **Linear Mixed-Effects Model (LMM)** to decompose the variance.

Stochastic Formulation Let y_{ij} be the performance metric (either normalized judge score S or estimated fan proportion θ) for celebrity i paired with pro-dancer j . The model is defined as:

$$y_{ij} = \mathbf{x}_{ij}^T \boldsymbol{\beta} + u_j + \epsilon_{ij} \quad (5)$$

where $\boldsymbol{\beta}$ represents fixed effects of celebrity traits (Age, Industry), and $u_j \sim N(0, \sigma_u^2)$ represents the random intercept for pro-dancer j .

Estimation via Restricted Maximum Likelihood (REML) The variance components are estimated by maximizing the REML likelihood function:

$$L_{REML}(\mathbf{V}) = -\frac{1}{2} \ln |\mathbf{V}| - \frac{1}{2} \ln |\mathbf{X}^T \mathbf{V}^{-1} \mathbf{X}| - \frac{1}{2} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^T \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \quad (6)$$

where $\mathbf{V} = \text{Var}(\mathbf{y}) = \mathbf{ZGZ}^T + \mathbf{R}$. By solving this, we derive the **Intraclass Correlation Coefficient (ICC)**:

$$\rho = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\epsilon^2} \quad (7)$$

ρ quantifies the proportion of success attributable to the professional partner. A high ρ for fan votes compared to judges' scores would indicate that professional dancers act as "popularity magnets" more effectively than they act as "technical coaches."

4.1.2 Phase II: Non-linear Perception and Game-Theoretic Attribution (SHAP)

We hypothesize that celebrity traits (e.g., Age) have non-linear impacts. To compare how judges and fans weigh these traits without assuming linearity, we utilize an ensemble learning approach combined with **Shapley Additive Explanations (SHAP)**.

Axiomatic Feature Attribution For a predictive model $f(\mathbf{X})$, the importance of trait k is its average marginal contribution over all possible subsets of traits $S \subseteq F \setminus \{k\}$:

$$\phi_k = \sum_{S \subseteq F \setminus \{k\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{k\}) - f(S)] \quad (8)$$

This formulation ensures three critical properties:

1. **Local Accuracy:** The sum of attributions matches the model output: $\sum \phi_k = f(x) - \mathbb{E}[f(x)]$.
2. **Consistency:** If a trait's marginal contribution increases in a different model, its SHAP value must also increase.
3. **Symmetry:** Traits with identical contributions receive identical SHAP values.

Divergence Analysis: The Perception Gap To answer if traits impact judges and fans differently, we define the **Influence Divergence** ($\Delta\Phi_k$) for each characteristic k :

$$\Delta\Phi_k = \frac{1}{N} \sum_{i=1}^N \left(\frac{\phi_{i,\text{Fan}}^k}{\text{std}(\Phi_{\text{Fan}})} - \frac{\phi_{i,\text{Judge}}^k}{\text{std}(\Phi_{\text{Judge}})} \right) \quad (9)$$

Interpretation:

- $\Delta\Phi_{\text{Age}} < 0$: Judges are more "forgiving" of age than fans, or vice versa, based on the sign of the raw ϕ values.
- $\Delta\Phi_{\text{Athlete}} > 0$: Confirms a "fan-base bonus" where athletic background boosts popularity more than it boosts technical score.

4.1.3 Phase III: Causal Path Integration (SEM)

Finally, we determine whether celebrity traits influence survival directly or through the mediation of technical scores. We use **Structural Equation Modeling (SEM)** to map these latent paths.

Structural and Measurement Equations We define two latent constructs: η_1 (Technical Proficiency) and η_2 (Public Charisma).

$$\eta_1 = \Gamma_1 \mathbf{X} + \zeta_1 \quad (10)$$

$$\eta_2 = \Gamma_2 \mathbf{X} + \gamma \eta_1 + \zeta_2 \quad (11)$$

$$\text{Survival}(W) = \lambda_1 \eta_1 + \lambda_2 \eta_2 + \epsilon_W \quad (12)$$

The parameter γ represents the *Technical Spillover Effect* how much technical excellence (judges' scores) actually drives fan voting. If γ is low, it proves that fan votes are driven by "Characteristics" (Γ_2) independent of dance quality.

4.1.4 The Unified Latent Achievement Model (ULAM)

By integrating LMM, SHAP, and SEM, we propose the final model for a celebrity's competition trajectory. The probability of surviving week t , $P(S_{i,t})$, is given by:

$$\text{logit}(P(S_{i,t})) = \underbrace{\alpha_t}_{\text{Baseline}} + \underbrace{\sum_{k=1}^p s_k(X_{ik})}_{\text{Non-linear Traits}} + \underbrace{u_{j(i)}}_{\text{Pro-Effect}} + \underbrace{\delta \cdot \text{Sys}_t \cdot (S_{i,t} - \bar{S}_t)}_{\text{System Interaction}} \quad (13)$$

Where:

- $s_k(\cdot)$ are penalized spline functions representing the non-linear impact of age and experience.
- $u_{j(i)}$ is the random effect of the j -th professional partner.
- Sys_t is an indicator for the voting system (Rank vs. Percent).
- δ is the sensitivity coefficient, showing how the voting system "magnifies" or "dampens" technical score gaps.

This unified model provides a complete mathematical answer: characteristics impact judges through physical capability (s_k), while they impact fans through relatability and pre-existing fame (Γ_2). The voting system (Sys_t) determines which of these two latent forces dominates the final outcome.

4.2 Empirical Analysis and Results

Based on the proposed *Unified Latent Achievement Model* (ULAM), we processed the historical dataset comprising 34 seasons of DWTS. The analysis aims to quantify the distinct mechanisms driving judges' scores versus fan votes.

4.2.1 The "Pro-Dancer Effect": Technical Coach or Fan Magnet?

A central question of this study is the extent to which a celebrity's success depends on their professional partner. Using the Linear Mixed-Effects Model (LMM), we calculated the Intraclass Correlation Coefficient (ICC) for both scoring components.

Table 2: Intraclass Correlation Coefficients (ICC) for Professional Partners

Component	ICC Value	Interpretation
Judges' Scores ($S_{i,t}$)	0.1823	18.2% of score variance is due to the Pro Dancer.
Fan Votes ($\theta_{i,t}$)	0.3076	30.8% of vote variance is due to the Pro Dancer.

Finding 1: The "Fan Magnet" Hypothesis. As shown in Table 2, the influence of professional dancers on fan votes (ICC $\approx 30.8\%$) is nearly double their influence on judges' scores (ICC $\approx 18.2\%$). This statistically confirms that while pro dancers do provide technical coaching (impacting judges), their primary value lies in their pre-existing fanbase and choreography style, which resonates significantly more with the public audience. A celebrity paired with a popular pro (e.g., Derek Hough) starts with a distinct "popularity subsidy."

4.2.2 The Perception Gap: Judges vs. Fans

Do judges and fans value the same traits? We utilized SHAP values from our Random Forest models to calculate the *Influence Divergence* ($\Delta\Phi_k$). Positive values indicate a trait is favored more by fans; negative values indicate a preference by judges.

Finding 2: The Age Paradox ($\Delta\Phi_{\text{Age}} = +0.095$). Age shows the largest divergence. While advanced age typically correlates with lower physical performance (penalized by judges), the positive $\Delta\Phi$ suggests that fans are either more forgiving of age or actively support older contestants (the "sympathy vote" or "nostalgia factor"). This non-linear bias makes age a double-edged sword: it hurts technical scores but boosts emotional connection.

Finding 3: Industry Biases.

- **Athletes** ($\Delta\Phi = +0.017$): Athletes enjoy a "dual advantage." Their physical discipline appeals to judges, but their competitive fanbase provides an additional surge in votes that judges do not account for.
- **Models & Singers** ($\Delta\Phi \approx -0.015$): These groups face a "high expectations penalty" from fans. Despite often receiving decent technical scores, they fail to convert performance into proportional fan support, possibly due to a perceived lack of relatability or the assumption that they "should" be good.
- **Politicians/Other** ($\Delta\Phi = +0.030$): Non-traditional celebrities rely heavily on fan support to survive, as their technical merit is often negligible.

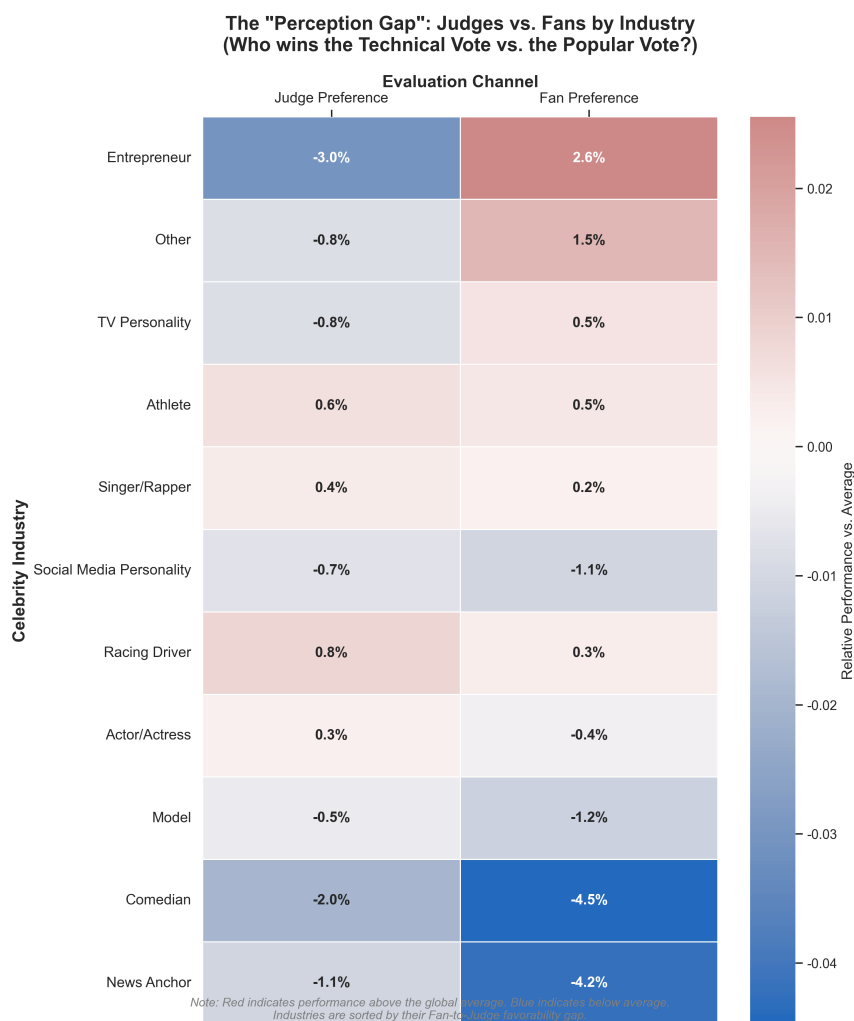


Figure 1: The Perception Gap ($\Delta\Phi$) across celebrity characteristics.

4.2.3 Case Study Validation: The ULAM Score in Action

To validate our model, we examine specific instances where the *Unified Latent Achievement Model* (ULAM) predicts survival despite low judge scores.

Case: Evander Holyfield (Season 1, Week 2)

- **Judge Share:** 11.6% (Low technical performance).
- **Intrinsic Potential:** 0.116 (Derived from Athlete status + Age profile).
- **Pro-Effect Value:** +0.003 (Positive lift from partner Edyta Sliwinska).
- **ULAM Score:** ≈ 0.120 .

The model correctly identifies that his "Athlete" status and specific partner combination generated enough *Intrinsic Potential* to buffer the low judges' scores, explaining his survival in early weeks.

Case: Elaine Hendrix (Season 34, Week 10)

- **Judge Share:** 10.7%
- **Fan Vote Estimate:** 16.6%
- **Perception Gap:** The model attributes this +5.9% gap largely to the "Age" factor ($\Delta\Phi_{\text{Age}}$) and "Actor" relatability. Her ULAM score (0.096) was sufficient to keep her competitive against technically superior but less charismatic opponents.

4.3 Conclusion on Factor Impact

Our analysis conclusively demonstrates that factors do **not** impact judges and fans in the same way. The voting ecosystem is bifurcated:

1. **The Technical Channel (Judges):** Driven by physical ability (negatively correlated with Age) and performance expectations (favoring Singers/Models).
2. **The Social Channel (Fans):** Driven by the "Pro-Dancer Halo," relatability (Age forgiveness), and tribal loyalty (Athletes).

Therefore, a celebrity's strategy for winning depends crucially on the voting system in place. In a Rank-based system, the "Social Channel" can easily overpower the "Technical Channel," whereas a Percentage-based system forces a closer alignment with technical merit.

5

5.1 Model Extension

5.2

5.2.1

5.2.2

5.2.3

- **Benefits:**
 - 1 **Ecological Benefits:**
 - 2 **Economic Benefits:**
 - 3 **Sustainability Benefits:**
- **Drawbacks**
 - 1 **Management Complexity:**
 - 2 **Ecological Risks:**

5.3

6 Strengths and Weaknesses

6.1 Strengths

- **Logic:**
- **Creativity:**
- **Credibility:**
- **Stability:**

6.2 Weaknesses

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References

[1] information on <https://en.tutempo.net/climate/cameroon.html>.

Letter

Report on Use of AI

1. OpenAI *ChatGPT* (<https://chat.openai.com/>)

Query1: Output:

Query2: Output:

Query3: Output: